

Name: Jianyu FU

Position: Ph.D. Student

Affiliation: Graduate School of Frontier Sciences,
The University of Tokyo

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Education

Ph.D. in Socio-Cultural Environmental Studies

Graduate School of Frontier Sciences, *The University of Tokyo*, Japan

2025 – Present

- Research area: Urban perception, mobility data analysis, and AI-based cityscape modeling.
- Supervisor: Prof. Yoshihide Sekimoto
- Member of JST SPRING GX program (Grant No. JPMJSP2108)

M.Arch. in Architecture

Southeast University (Project 985), Nanjing, China

2021.09 – 2024.06

- Grade average: **90.1 / 100 (Rank 1 / 36)**
- Research direction: Urban Architecture, Urban Planning and Design, Urban Analytics, and Big Data Analysis.

B.Arch. in Architecture

Northeastern University (Project 985), Shenyang, China

2015.09 – 2020.06

- Undergraduate thesis focused on sustainable architectural design and human-environment interaction.

Publications & Conference Papers

- **FU, J. (2025).** *Comprehensive Evaluation Method of Rooftop Views in High-Density Urban Areas: A Case Study of Guangzhou's Historic Districts.*
Urban Design International (Springer Nature).
- **FU, J. (2025).** *Urban Real Accessibility Evaluation Based on GPS Data: Integration of Space Syntax and Deep Learning.*
Poster accepted for presentation at **NetMob 2025**, MIT & Inria, France.
- **FU, J. (2025).** Linking Urban Morphology and Human Activity Patterns.
IEEE BigData 2025. Oral presentation
- **FU, J. (2024).** **Research on Sustainable Renewal of Historic Blocks Based on Urban Catalyst Theory.**
Applied and Computational Engineering
- **FU, J. (2022, November).** *Research on Urban Public Space Quality Evaluation Model Based on Scene Photo's Deep Learning.*
the 2022 Computational Design Symposium and Annual Conference of the Computational Design Academic Committee of the Architectural Society of China.

- **FU, J.** (2022, November). *Spatial Cognitive Preference of Traditional Village Nodes from the Perspective of Eye-Tracking Technology*.
the 2022 Computational Design Symposium and Annual Conference of the Computational Design Academic Committee of the Architectural Society of China.
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Research Interests

- Urban perception and emotion modeling
 - Cityscape analytics and environmental cognition
 - AI and deep learning in urban morphology
 - Big data analysis for urban design and planning
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Technical Skills

- **Programming:** Python (3+ years),
 - **Machine Learning & Deep Learning:** PyTorch, TensorFlow, scikit-learn, Keras
 - **Spatial Analysis & GIS:** QGIS, ArcGIS, Kepler.gl, PLATEAU 3D City Model
 - **Data Processing:** Pandas, GeoPandas, OpenStreetMap & social media APIs
 - **Visualization:** matplotlib, seaborn, plotly, Power BI
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Professional Experience

Sponsored Researcher

SOMAR at ICPSR, University of Michigan

2025 – Present

- Appointed as a sponsored researcher/member of the University of Michigan with access to U-M computing services.
- Granted approved access to Meta Content Library API data through the SOMAR Virtual Data Enclave (VDE) under SOMARAPPLY-768 (SOMAR Data Application).
- Conduct research using publicly available Facebook and Instagram social media data in a secure computing environment for urban perception and collective emotion analysis.

Research Assistant

Sekimoto Laboratory, Center for Spatial Information Science (CSIS), The University of Tokyo

2025 – Present

- Conducted multimodal data analysis on urban perception and mobility behavior.
 - Processed large-scale GPS, SNS, and image datasets under strict privacy protocols.
 - Participated in SPRING GX interdisciplinary research on sustainable, emotion-aware cities.
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Awards & Grants

- JST SPRING GX Fellowship, The University of Tokyo (2023–Present)
- NetMob 2025 Poster Track Selection (2025)
- Best Presentation Award at IEEE BigData 2025